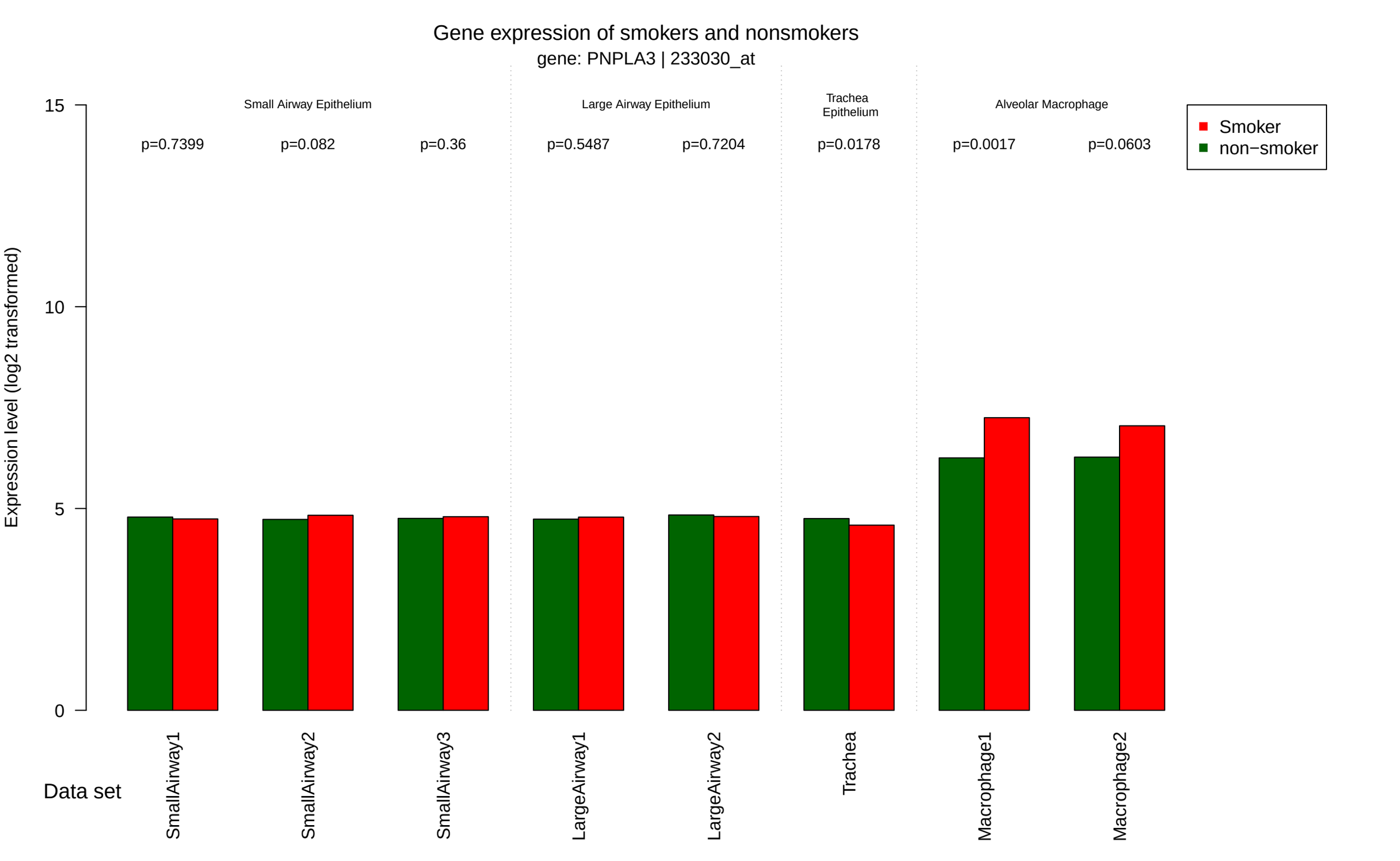
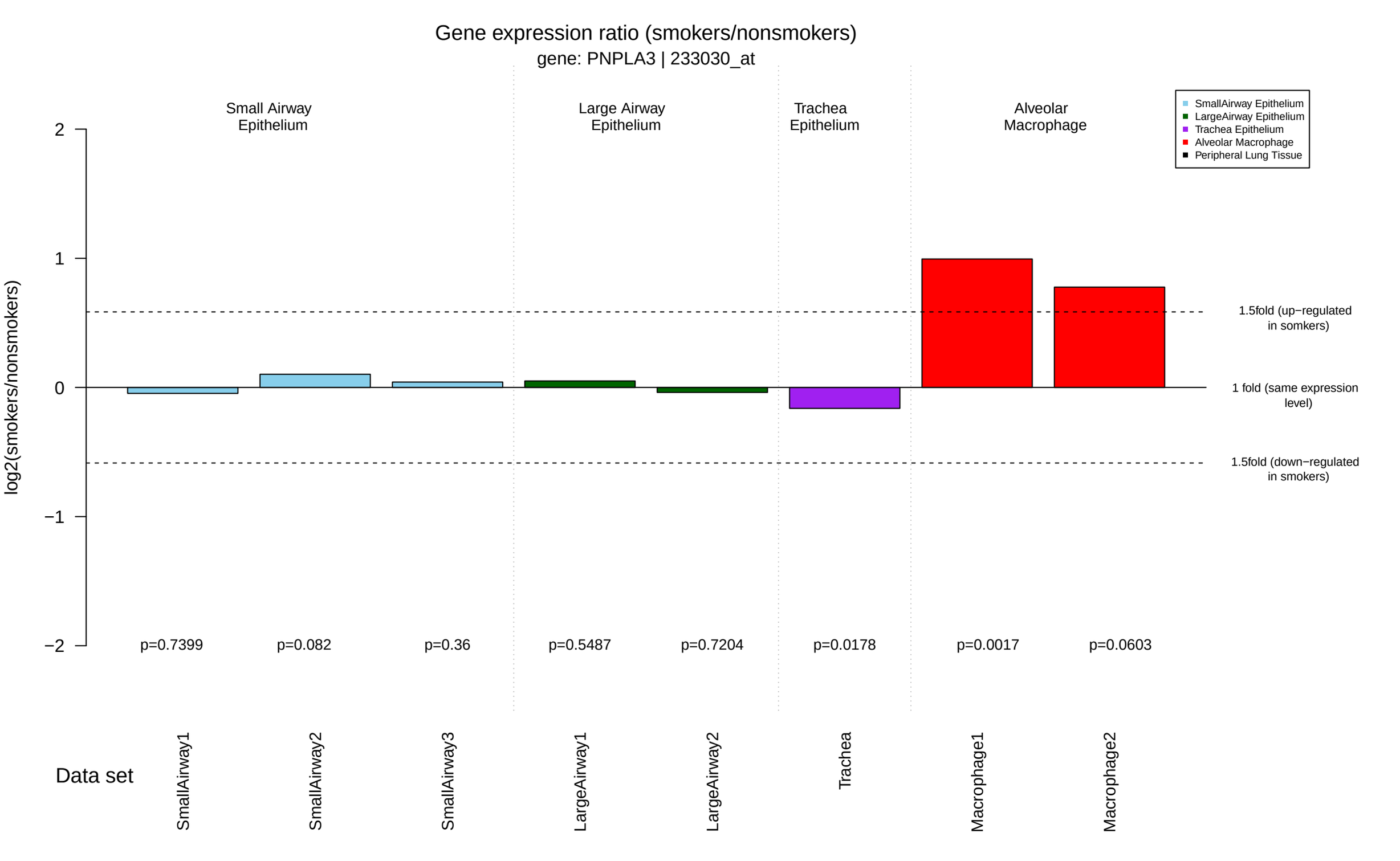
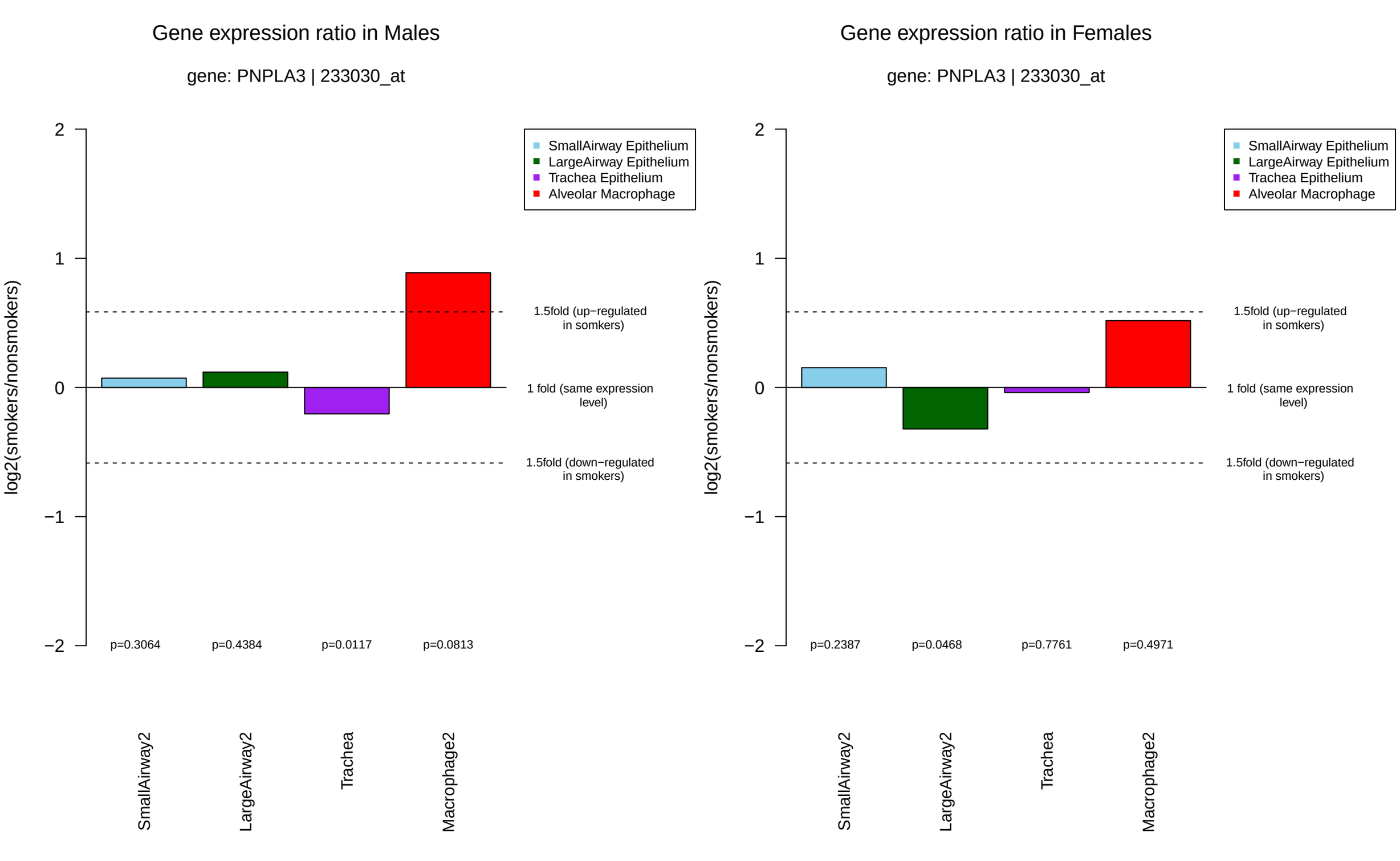
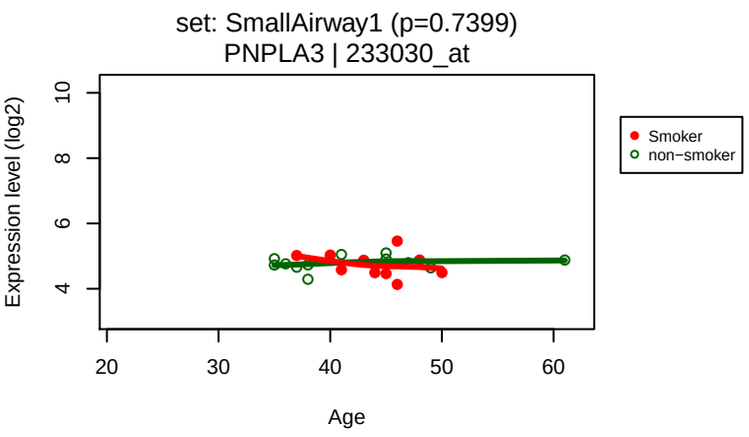
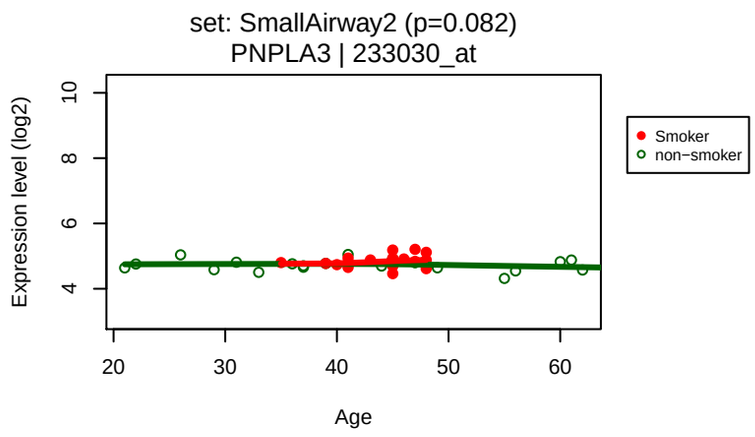
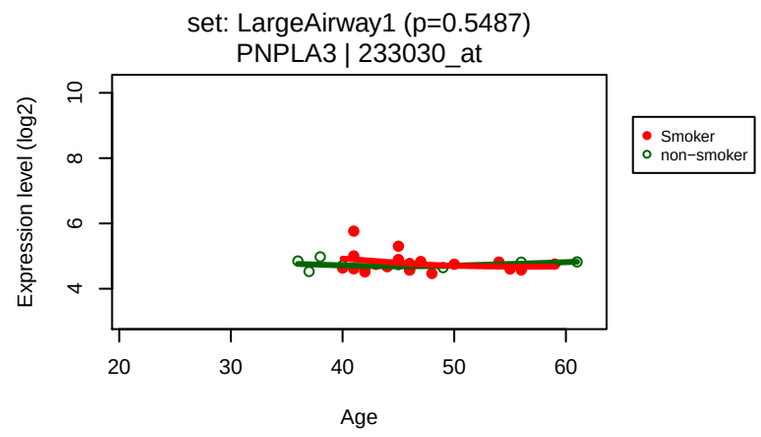

Gene expression of smokers and nonsmokers
gene: PNPLA3 | 233030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7399
p=0.082
p=0.36
p=0.5487
p=0.7204
p=0.0178
p=0.0017
p=0.0603
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

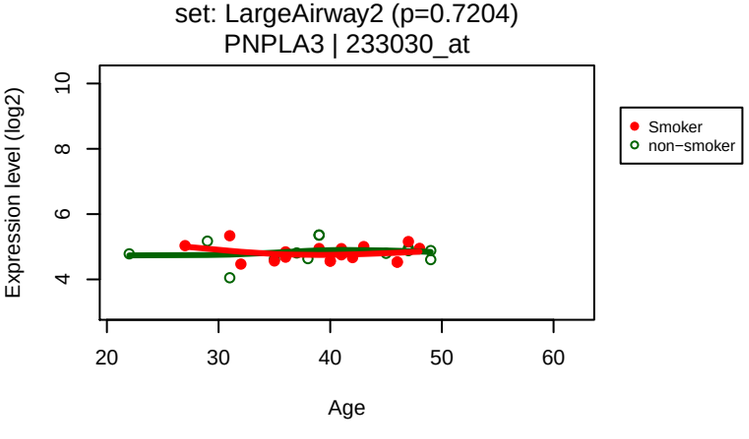

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA3 | 233030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7399
p=0.082
p=0.36
p=0.5487
p=0.7204
p=0.0178
p=0.0017
p=0.0603
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

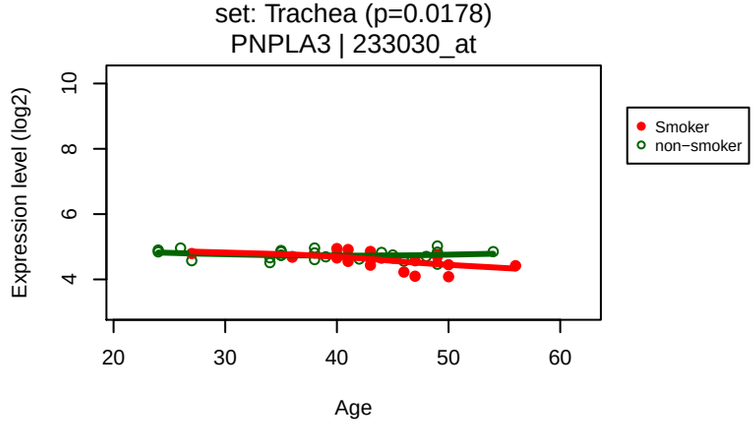

Gene expression ratio in Males
gene: PNPLA3 | 233030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3064
p=0.4384
p=0.0117
p=0.0813
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA3 | 233030_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2387
p=0.0468
p=0.7761
p=0.4971
SmallAirway2
LargeAirway2
Trachea
Macrophage2

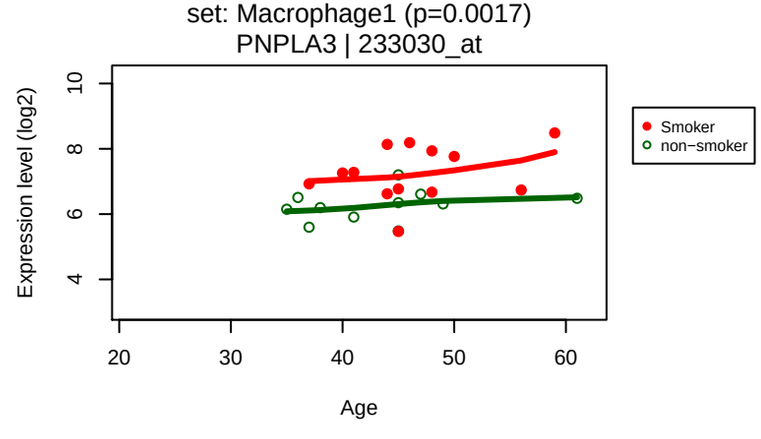

set: SmallAirway1 (p=0.7399)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker

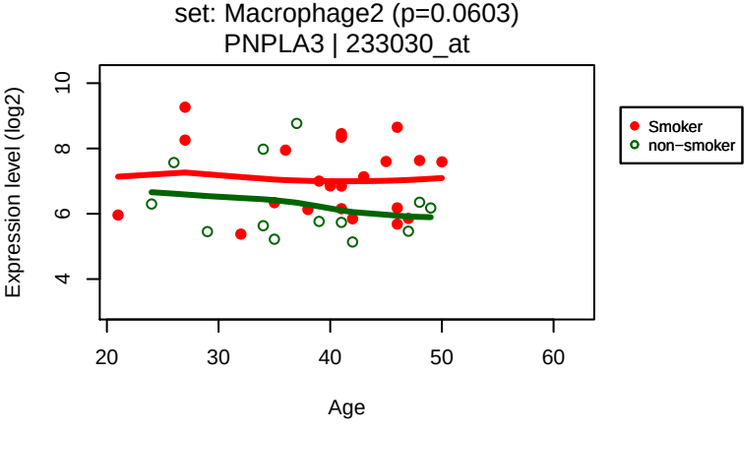

set: SmallAirway2 (p=0.082)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker

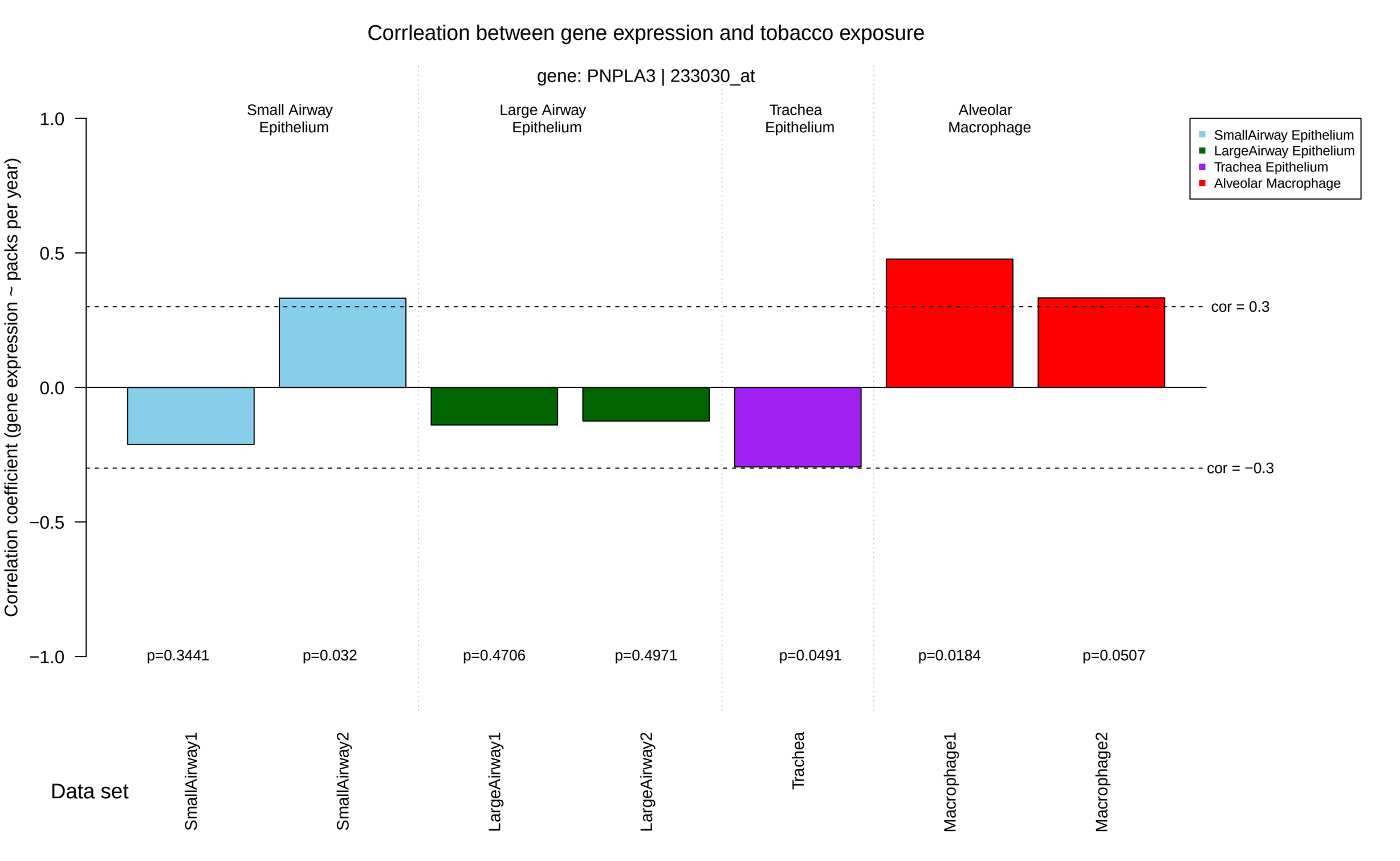

set: LargeAirway1 (p=0.5487)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7204)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0178)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0017)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0603)
PNPLA3 | 233030_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA3 | 233030_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3441
p=0.032
p=0.4706
p=0.4971
p=0.0491
p=0.0184
p=0.0507
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2